Khadijah Aamir Chaudhry

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LinkedIn · GitHub

EDUCATION

Lahore University of Management Sciences (LUMS) *2023–2027*
BSc Computer Science · Minor in Computer Engineering

Lahore Grammar School *2018–2023*
A Levels: 4 As · O Levels: 9 A*s · SAT: 1530

RESEARCH INTERESTS

Secure embedded systems · Hardware roots of trust · Hardware–software co-design · Trusted execution · Applied cryptography

RESEARCH EXPERIENCE

Research Assistant, CHASE Lab, LUMS *Aug 2025–present*
Supervisor: Dr. Muhammad Ali Siddiqi
- Benchmarked post-quantum algorithms on ESP32 and built a **simulated RISC-V trace testbed** for on-chip monitoring.

Research Assistant, SysNet Lab, LUMS *Jan–May 2026*
Supervisor: Dr. Naveed Anwar Bhatti
- Co-developed **EyeSense's smartphone ToF/ML pipeline** and helped collect eyesight data in lab and eye-camp settings.

SELECTED PROJECTS

On Chip Behavioral Monitoring *2026–present*
- Built a Rocket Chip/Verilator testbed with four RV32 configurations and **TileLink MMIO trace capture**; replayed benign and trigger-active firmware.

Protected Media Containers *2026–present*
- Developed a system design for **protected camera/audio capture**, encrypted storage, display, and sharing under an untrusted mobile OS, informed by TEE and DRM research.

EyeSense: Mobile Eyesight Screening *Jan–May 2026*
- Co-developed a smartphone ToF phase-map and machine-learning pipeline for refractive-error screening. The team collected data from 120 participants and reported **1.15 D mean absolute error**.

Post-Quantum Cryptography on ESP32 *Aug–Dec 2025*
- Benchmarked **Kyber-512, Light-SABER, and Dilithium-II** on ESP32 for latency, memory, network cost, and estimated energy use.

TEACHING EXPERIENCE

Teaching Assistant, CS100, LUMS *Aug 2026–present*
- Supported about **450 first-year students**: designed and conducted labs, prepared and graded quizzes, held weekly office hours, and invigilated exams.

PROFESSIONAL EXPERIENCE

Marketing Manager, Fintelligent Solutions *Jul 2026–present*
- Coordinated the team delivery of **brand identity and company website**, plus office collateral and LinkedIn launch content for a Microsoft Dynamics 365 consultancy.

Data Collection Coach, Uplift AI *Jul–Aug 2024*
- Led **10 data collectors** for a Pakistani-language AI dataset; ran daily quality reviews, feedback sessions, and field outreach.

LEADERSHIP & SERVICE

Peer Ambassador Leader, LUMS *Aug 2026–present*
- Guided **15 incoming students** through orientation and faculty introductions; continuing monthly first-year support sessions.

Women in STEM Coordinator, LUMS SPADES *Sep 2024–May 2025*
- Helped launch the **first women-only STEM event at PSIFI** for more than 100 participants.

Co-Founder and Logistics Director, Umeed-e-Nau *Sep 2021–May 2023*
- Organized **inclusive educational activities** and visits for underprivileged and disabled children.

SELECTED GRADUATE COURSEWORK

Internet of Things · Applied Cryptography · Topics in Computer and Network Security · Distributed Systems

HONORS AND AWARDS

- **100% Merit Scholarship**, Lahore Grammar School *Aug 2021–May 2023*
- **Shortlisted writing submission**, LUMSx Nexus *May 2025*
- **Honorable Mention**, EducationUSA Pakistan Writing Contest *Sep 2022*

TECHNICAL SKILLS

Programming languages: C · C++ · Python · JavaScript
Embedded and hardware: ESP32 · RISC-V · Rocket Chip · Verilator · FreeRTOS
Machine learning and sensing: ToF sensing · CNNs · ridge regression
Tools and methods: Docker · Git · post-quantum cryptography benchmarking